

SNA-SKAN: Unpaired learning for SDOCT speckle noise removal based on self noise assist and kolmogorov-arnold network

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ABSTRACT

Optical Coherence Tomography (OCT) will inevitably be contaminated by speckle noise when imaging, resulting in a decrease in the visual quality of images and affecting clinical diagnosis. Existing unsupervised denoising methods often rely on complex model architectures or extensive data preprocessing. This paper proposes an unpaired Spectral-Domain OCT (SDOCT) denoising framework named SNA-SKAN. The Self Noise Assist (SNA) module leverages wavelet transform and singular value decomposition to extract noise components directly from noisy OCT images. These components are then fused into a new noise representation, which guides the neural network in effectively learning speckle noise patterns. Furthermore, to more effectively model speckle noise in OCT images, this paper exploits the Kolmogorov-Arnold Network (KAN) for its superior capacity to represent complex distributions, and proposes a KAN-based speckle noise generation network (SKAN). The SNA-SKAN framework is built upon the Generative Adversarial Network (GAN) architecture, employing a single generator and a single discriminator. Extensive experiments conducted on an unpaired public dataset for training and two public datasets for evaluation demonstrate that the proposed method outperforms existing unsupervised methods and state-of-the-art unpaired methods, in terms of denoising capability and detail preservation. SNA-SKAN can achieve efficient OCT denoising while preserving edges and details, demonstrating strong potential to meet clinical needs. The code is publicly available at: <https://github.com/zhencunjiang/SNA-SKAN>.

1. Introduction

Optical Coherence Tomography (OCT) is a non-invasive imaging method based on the principle of low-coherence light interference, which utilizes the differential reflection of light by different tissues to perform two-dimensional or three-dimensional imaging. OCT is widely employed in clinical diagnostics, particularly in ophthalmology and dermatology (Akpinar et al., 2024; Achim et al., 2023). However, due to the highly scattering nature of biological tissues, OCT images are inevitably affected by speckle noise, which degrades visual quality and hinders accurate diagnosis (Pereg, 2024; Darlow et al., 2015). The mainstream OCT denoising methods can be categorized into two types, hardware improvement-based and image processing-based. While enhancing hardware components can reduce imaging noise, it often entails increased system complexity and higher costs. As a result, recent research has increasingly focused on image processing methods for speckle noise reduction. The image processing-based denoising methods

can be further categorized into traditional methods and deep learning-based methods. Traditional methods, such as Block-Matching 3D (BM3D) (Chong and Zhu, 2013; Danielyan et al., 2011) and Non-Local Means (NLM) (Aum et al., 2015; Buades et al., 2011), can attenuate speckle noise to a certain extent, however, they often leave residual noise and may lead to the loss of fine structural details. In contrast, deep learning-based denoising methods have shown significant progress in recent years. Compared with traditional approaches, these methods can achieve more effective noise suppression while better preserving image details.

When paired data are available, supervised learning-based methods can be applied for denoising and often yield excellent performance. However, obtaining high-quality paired data is a stringent condition. Typically, clean OCT images are obtained through techniques such as multi-frame averaging, which is time-consuming. During the scanning process, motion artifacts frequently occur, leading to detail loss or structural deformation in the final averaged results (Guo et al., 2019a;

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Huang et al., 2020a). In scenarios where obtaining large paired dataset is infeasible, utilizing unpaired clean-noisy images for unsupervised learning-based OCT denoising are more significant. Some unsupervised learning methods attempt to exploit the intrinsic information within noisy images for denoising. They consider the background noise in OCT images as pure noise and assume that its distribution is consistent with the noise distribution in the detail regions of OCT images. Thus, the background noise is used to guide the denoising process across the entire OCT image (Guo et al., 2020). In practical applications, however, this method faces two major limitations. First, the assumption that background noise is purely isolated cannot be reliably guaranteed. Second, manually distinguishing background from detailed regions is both time-consuming and labor-intensive.

Unpaired methods utilize independent clean and noisy OCT images for training. Currently, most unpaired OCT denoising methods are built upon the CycleGAN framework (Zhu et al., 2017; Wu et al., 2021a). However, this framework requires two generators and two discriminators, which make it extremely challenging to train. Mu et al. argue that the CycleGAN framework is overly complex and redundant for OCT denoising tasks. To address this issue, they proposed an unpaired denoising method called Triplet Cross-Fusion Learning (TCFL), which employs only one generator and one discriminator (Geng et al., 2022). This method achieves OCT denoising by training on three unpaired images (two noisy images and one clean image). However, this method also has several limitations. It relies entirely on neural networks to extract noise, which makes it difficult to accurately capture the underlying mathematical distribution of speckle noise. As a result, this may lead to incomplete noise removal, unintended loss of structural information, and underutilization of the intrinsic features present in noisy images. In addition, this method requires three unpaired images for training, imposing high demands on dataset size. Recently, Kolmogorov-Arnold Networks (KANs) have demonstrated superior

capability in approximating complex functions and have shown robust performance across various applications. Considering that speckle noise follows a complex mathematical distribution, it makes sense to effectively leverage KAN to generate speckle noise.

To address the limitations of existing OCT denoising methods, this paper proposes an unpaired OCT denoising method based on Generative Adversarial Network (GAN), which consists of the Self Noise Assist (SNA) module and KAN-based speckle noise generator (SKAN). Specifically, the SNA module utilizes wavelet transform and Singular Value Decomposition (SVD) to extract high-confidence speckle noise from noisy OCT images. This extracted noise will be used to guide the neural network to learn the mathematical distribution of speckle noise. The SNA module allows two types of noise to interact and fuse dynamically, ensuring that the noise representation remains flexible rather than fixed. Furthermore, patch shuffle is employed to disrupt the structural information in the two types noise extracted by SNA, thereby preventing the loss of meaningful structural features during denoising. SKAN leverages the KAN's capacity to model complex distributions, enabling more realistic speckle noise generation and, consequently, more effective noise removal. The main contributions of this paper are summarized as follows:

- 1) An unpaired Spectral-Domain OCT (SDOCT) denoising method SNA-SKAN is proposed, which eliminates the need for complex data preprocessing and network architectures.
- 2) The Self Noise Assist (SNA) module is introduced to directly learn the distribution of speckle noise, alleviating the challenges of neural network training. It utilizes traditional denoising methods to extract high-confidence speckle noise and enhances noise diversity through noise dynamic weighting and fusion, thereby guiding neural network training more effectively.

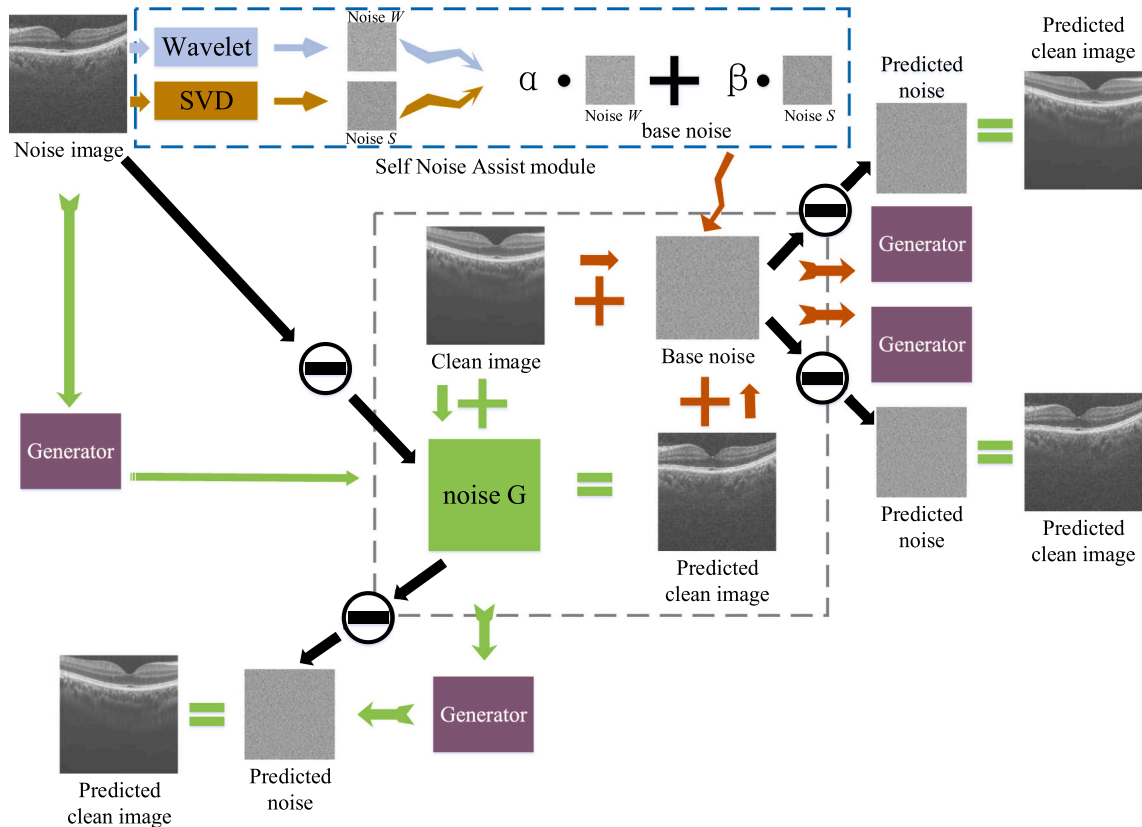


Fig. 1. The overview of the SNA-SKAN. The green and brown swallow-tailed arrow indicates the noisy image input to the generator, the brown arrow indicates adding base noise to the clean OCT, and the green arrow indicates adding noise G to the OCT.

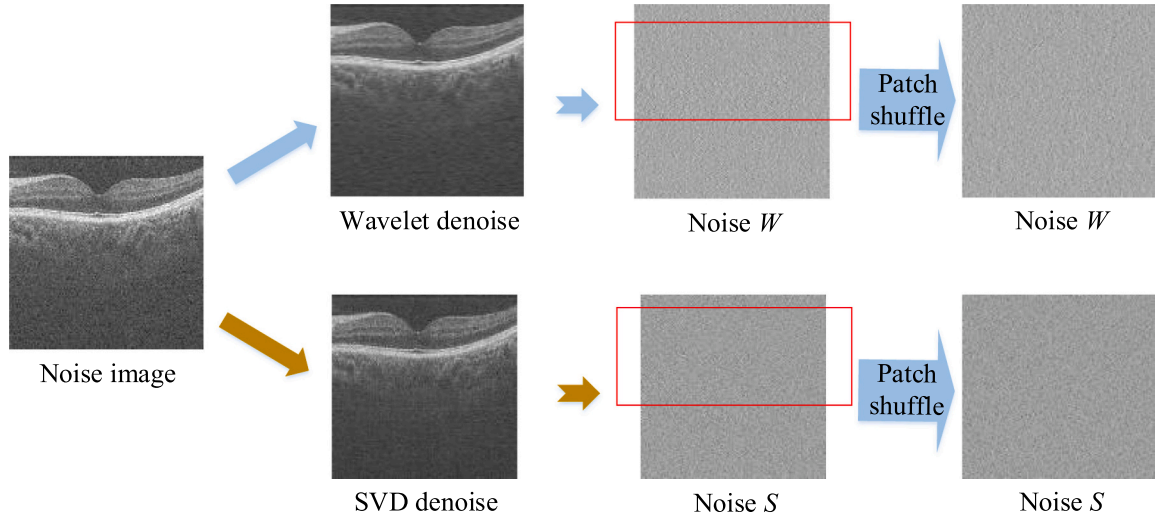


Fig. 2. The illustration for noise extraction in SNA. The red box indicates the structural information contained in the extracted speckle noise.

- 3) A KAN-based speckle noise generation network SKAN is proposed, which leverages the representational power of KANs to model complex distributions, enabling more accurate and diverse speckle noise generation.

2. Related work

2.1. Traditional and deep learning combined denoising methods

In recent years, numerous studies have explored the integration of traditional image processing methods with deep learning methods for medical image denoising. Chen et al. proposed combining fractional-order filtering and boosted singular value shrinkage to remove speckle noise from OCT images (Chen et al., 2019). Fang et al. proposed a method based on singular value decomposition and fractional edge detection to remove speckle noise from OCT images (Fang et al., 2023). Yeganli et al. proposed a method of iterative regularization based on higher-order singular value decomposition to remove noise from MRI images (Yeganli et al., 2017). Liu et al. proposed a method based on weighted higher-order singular value decomposition to achieve denoising of 3D PET images (Liu et al., 2016). Zhou et al. proposed a mini-cGAN and integrated it into the CycleGAN framework to achieve speckle noise removal in OCT images (Zhou et al., 2021). Eunhee Kang et al. proposed a frame-based wavelet residual neural network to denoise LDCT images (Kang et al., 2018). Gu et al. combined wavelet transform with CycleGAN to achieve unsupervised denoising of low-dose CT images (Gu et al., 2021). Mamta et al. proposed a Bayesian shrinkage-based wavelet transform and block-based autoencoder to denoise MRI images (Juneja et al., 2021). Gadipudi et al. proposed a multi-scale residual grouped convolutional neural network based on wavelet transform to achieve denoising of CT images (Amaranageswarao et al., 2020). Sourya et al. proposed a residual edge-aware GAN based on wavelet transform to achieve denoising of OCT images (Sengupta et al., 2021). Hee et al. proposed a GAN framework based on batch normalization and cascade of wavelet residual channel attention blocks to reduce speckle noise in ultrasound images (Khor et al., 2022). From the above literature, it can be inferred that combining traditional methods with deep learning methods can improve the performance of OCT denoising.

2.2. Deep learning-based OCT denoising methods

Deep learning-based methods have become the predominant approach across various fields (Cantone et al., 2024; Flaborea et al.,

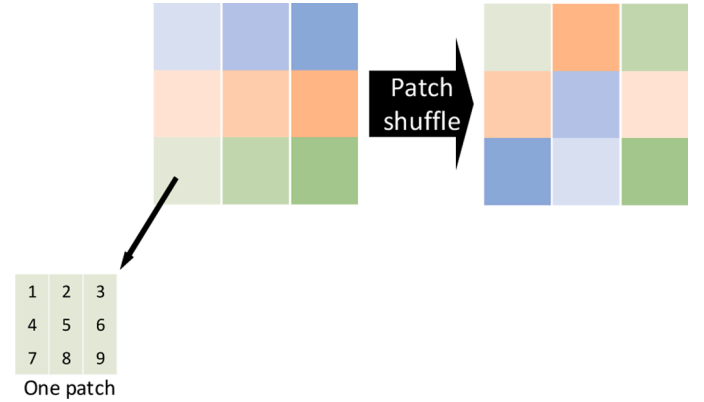


Fig. 3. Illustration of Patch shuffle.

2024; Rofena et al., 2024; Tortora et al., 2021). Deep learning-based OCT denoising methods can be primarily categorized into supervised learning-based methods and unsupervised learning-based methods. Supervised learning-based methods require paired clean-noisy OCT images for training, and the model structures are mostly CNN-based or transformer-based. These methods typically employ L_1 loss or uniquely designed loss functions for training. Xu et al. proposed OCTNet, a convolutional neural network incorporating residual learning, to achieve OCT denoising (Xu et al., 2020). Quan Zhou et al. introduced a model that integrates cross-scale CNN with an intra-patch and inter-patch based transformer, leading to successful OCT denoising (Zhou et al., 2022). Sheng et al. proposed a double-path parallel CNN to achieve deeper-level extraction of speckle noise (Shen et al., 2021). Ma et al. proposed an edge-sensitive loss function to preserve clearer edges while denoising OCT images (Ma et al., 2018). Qiu et al. proposed a perceptually-sensitive loss function to preserve more details in OCT denoising (Qiu et al., 2020). Some unsupervised learning-based methods divide noisy OCT images into background and detail regions, leveraging the intrinsic information from the background regions for denoising. However, this requires a lot of time for data preprocessing. Most of unpaired OCT denoising methods are mainly implemented from the perspective of CycleGAN framework. Wu et al. used the CycleGAN framework to convert SD-OCT images into high-quality enhanced depth imaging OCT images to achieve OCT denoising (Wu et al., 2021b). Guo et al. incorporated structural similarity loss into the loss function under CycleGAN framework to achieve OCT image denoising while preserving more structural information (Guo et al., 2019b). Huang et al. combined

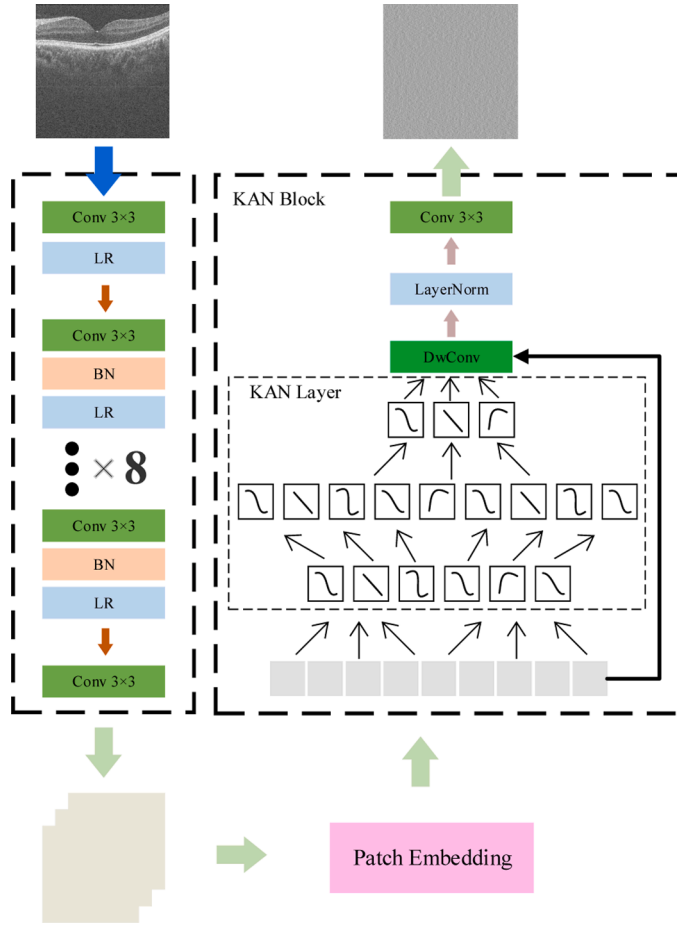


Fig. 4. The structure of the generator SKAN.

noise-powered disentangled representation with the CycleGAN framework to reduce speckle noise in OCT images (Huang et al., 2020b). Manakov et al. proposed a framework of HDcycleGAN, in which hybrid discriminator is employed, the two discriminators share the same parameters, and skip connection is introduced into the generator. HDcycleGAN achieves discrimination of subtle variations of noise on OCT images (Manakov et al., 2019). Supervised learning-based methods requires paired clean-noisy OCT images, but unsupervised learning-based methods don't. What unsupervised learning-based methods utilize is the intrinsic information of noisy OCT images, however they require extensive data preprocessing. The existing methods dealing with unpaired OCT images do not fully explore the intrinsic information of OCT images, which may lead to a risk of complex methodology framework.

2.3. Kolmogorov-arnold networks

KAN is a novel neural network architecture inspired by the Kolmogorov-Arnold representation theorem (Liu et al., 2024). Li et al. integrated it into UNet for segmentation tasks and as a noise predictor in diffusion models (Li et al., 2024). Tao et al. applied a KAN-based diffusion model for MRI reconstruction (Tao et al., 2025). Li et al.

Table 1

The details of these three OCT datasets.

	PKU37	DUKE17	DUKE28
Scanner	SDOCT	SDOCT	SDOCT
Noisy image	1734	17	28
Clean image	37	17	28
Image size	640 × 640	520 × 970	450 × 900
Train/test	Train	Test	Test

utilized a KAN-based diffusion model for hyperspectral and multispectral image fusion (Li et al., 2025). These studies demonstrate that KAN can be effectively used to model complex noise distributions.

The proposed SNA-SKAN method eliminates the need for extensive preprocessing and effectively leverages the intrinsic information of noisy OCT images, as well as KAN's strong capability for modeling speckle noise.

3. SNA-SKAN method

3.1. Motivation

The goal of deep learning-based OCT denoising methods is to train a neural network that can accurately extract noise or generate noise-free OCT images. Accordingly, the core challenge lies in how to effectively train such a neural network. TCFL (Geng et al., 2022) departed from the CycleGAN framework and achieved unpaired OCT denoising with single generator and single discriminator. However, the noise on noisy images is directly extracted by the generator, due to lack of paired clean-noisy images, the training results may deviate from the correct values. Thereafter, we are motivated by such an idea that, if high confidence speckle noise is involved in and guide the training process, the probability of the network training progressing towards the correct direction will increase. This is because the intrinsic information of the noisy OCT images is fully utilized.

3.2. OCT speckle noise model

In general, speckle noise in OCT is considered multiplicative noise. However, in clinical practice, SDOCT data are typically projected into logarithmic space for displaying images. This transformation alters the speckle noise characteristics, effectively converting it from multiplicative to additive noise (Aumann et al., 2019; Salinas and Fernández, 2007). Consequently, the noisy SDOCT images NI can be defined as the sum of clean OCT images CI and speckle noise N , as shown in formula (1).

$$NI = CI + N \quad (1)$$

3.3. SNA-SKAN method overview

The proposed method, SNA-SKAN, is illustrated in Fig. 1. This method is based on a GAN framework consisting of a single generator and a single discriminator. First, the noisy OCT images are fed into the SNA module, where they undergo wavelet transform and singular value decomposition (SVD), resulting in two types of extracted noise, W and S . Next, noise W and S are fused to form a composite base noise. Meanwhile, the noisy OCT images are fed into the generator, and the noise G and the predicted clean OCT image will be output. The clean OCT image is added to the base noise and noise G , respectively, and the predicted clean OCT is combined with the base noise. Then these new noise images obtained are input into the generator for further training to generate noise.

3.4. SNA module

SNA is a module consisting of traditional denoising methods. Traditional methods typically rely on mathematical transformations and thresholding to extract noise and do not require paired clean-noisy images for training, thereby allowing effective utilization of the intrinsic information within the images. Traditional methods are employed to extract high confidence speckle noise from the noisy OCT images, which are then used to guide the neural network in extracting speckle noise. SNA employ wavelet transform and SVD as the noise extractor. SVD is a linear algebra-based image denoising method. It decomposes the image matrix into singular values and corresponding singular vectors, filters

Table 2

Test results on Duke17.

Model	PSNR↑	SSIM↑	RMSE↓
Noisy	16.273	0.311	0.156
BM3D (Danielyan et al., 2011)	18.888	0.811	0.119
NLM (Buades et al., 2011)	18.351	0.696	0.125
DDPM (Hu et al., 2022)	16.432	0.306	0.152
CycleGAN (Gu et al., 2021)	17.991	0.785	0.129
TCFL (Geng et al., 2022)	26.085	0.776	0.050
SNA-SKAN	26.651	0.814	0.046

Table 3

Test results on Duke28.

Model	PSNR↑	SSIM↑	RMSE↓
Noisy	16.599	0.311	0.149
BM3D (Danielyan et al., 2011)	18.626	0.804	0.121
NLM (Buades et al., 2011)	18.585	0.776	0.122
DDPM (Hu et al., 2022)	16.603	0.312	0.148
CycleGAN (Gu et al., 2021)	17.302	0.774	0.141
TCFL (Geng et al., 2022)	27.199	0.791	0.045
SNA-SKAN	27.844	0.827	0.040

out the smaller singular values associated with noise, and reconstructs the image using the remaining components to complete the denoising process (Andrews and Patterson, 1976). Wavelet transform decomposes the original image into multiple frequency components, where noise

typically manifests in the high-frequency bands (Chang et al., 2000). The frequency components are removed by prespecified thresholds, followed by inverse wavelet transform, and then the image is reconstructed to obtain the denoised image. The noise extracted by SVD and wavelet transform are shown in Fig. 2.

The red box in Fig. 2 highlights that noise W and S inevitably contain some structural information. If this structural information is used to guide the neural network, the model may mistakenly interpret it as noise, potentially resulting in the unintended removal of meaningful structures during denoising. If all pixels on the noisy OCT image are shuffled randomly, the inter-pixel correlations on the noisy OCT image will be lost. Therefore, patch shuffle is employed to disrupt the structural information in the noise in order to preserve the correlations between pixels. Patch shuffle involves dividing a larger image matrix into several smaller patches and randomly shuffling them, which disrupts global structure while retaining local statistical properties, as shown in Fig. 3.

If the auxiliary noise remains fixed, the neural network will tend to learn a combination of the noise extracted via wavelet transform and SVD. In this case, the optimal state of this neural network can only extract the combined noise, it will be unable to become a more accurate noise removal method. To address this, this paper introduces two adjustable coefficients to control the noise added to the clean OCT image. The SNA obtains noise W through wavelet transform and noise S through SVD. Adding adjustable coefficients α and β to these two types of noise, as (2).

$$\text{noise base} = \alpha * \text{noiseS} + \beta * \text{noiseW} \quad (2)$$

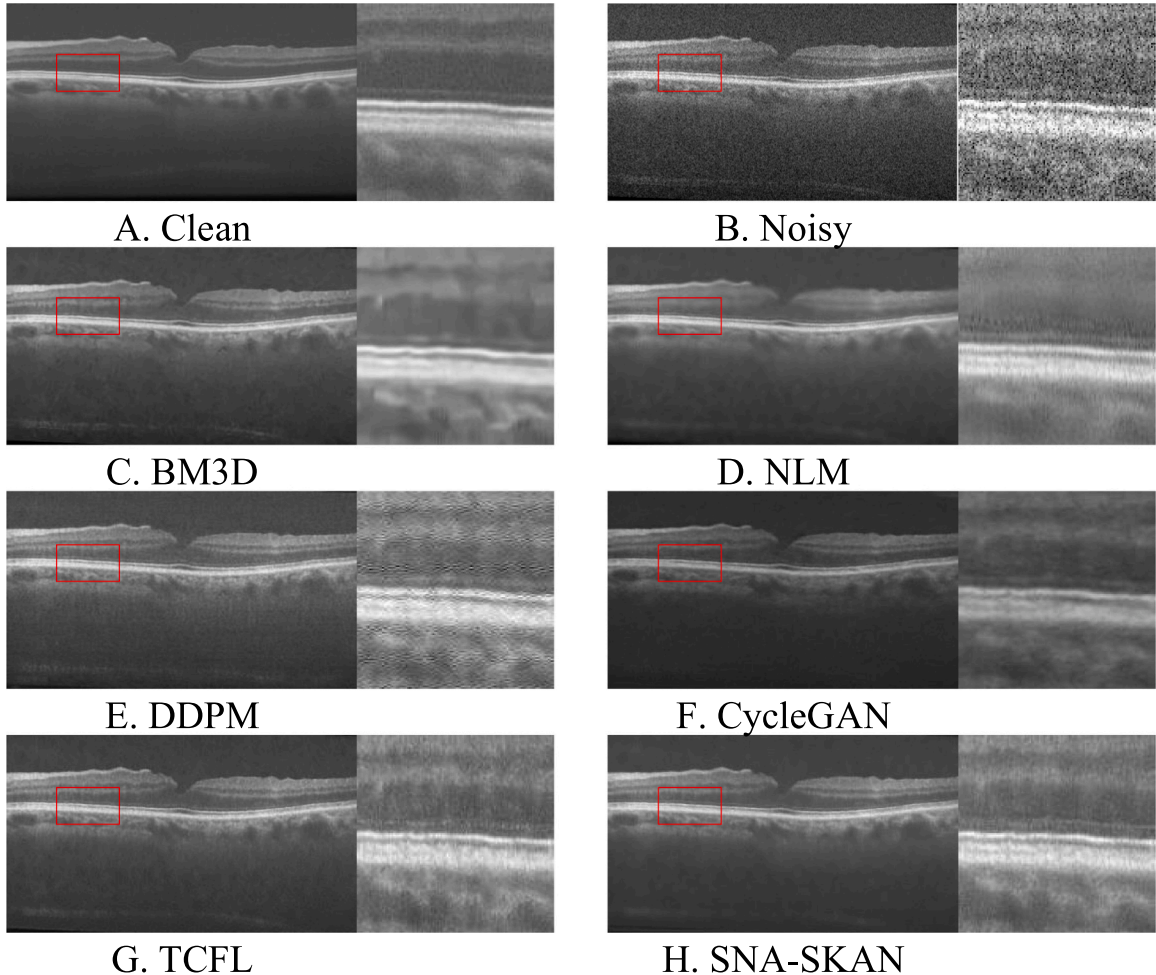


Fig. 5. The denoising results on Duke17. The left images of each image pair display clean or noisy or denoised OCT images, the right images of each image pair display concrete views of red box regions.

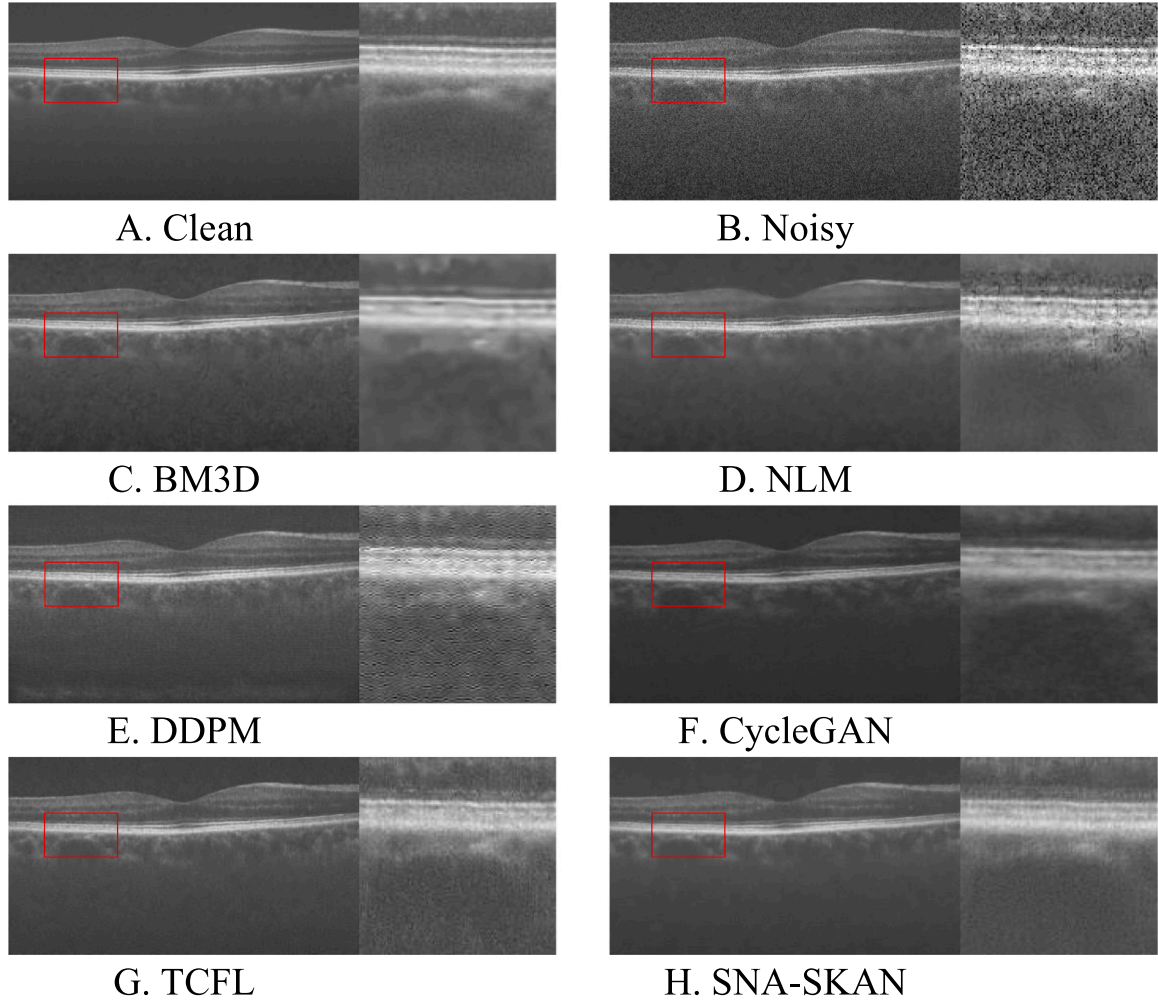


Fig. 6. The denoising results on Duke28. The left images of each image pair display clean or noisy or denoised OCT images, the right images of each image pair display concrete views of red box regions.

Table 4

The comparison of model parameters and running time.

Method	Parameters (Mbyte)	Inference time(s)
BM3D (Danielyan et al., 2011)	-	1.976
NLM (Buades et al., 2011)	-	0.723
DDPM (Hu et al., 2022)	8.33	1.561
CycleGAN (Gu et al., 2021)	173.36	0.128
TCFL (Geng et al., 2022)	1.13	0.076
SNA-SKAN	1.33	0.116

3.5. SKAN

The proposed speckle noise generation network, SKAN, consists of a feature extraction module and a noise generation module. The feature extraction module consists of several convolutional layers, while the noise generation module is implemented based on KAN, incorporating a

Table 5

Test results on different noise intensity.

Variance levels	PSNR↑	SSIM↑	RMSE↓
0.02	30.549	0.893	0.029
0.05	29.918	0.871	0.032
0.1	29.162	0.846	0.034
0.2	27.995	0.806	0.040

linear projection layer followed by a KAN block. The KAN block comprises three KAN layers, a depth-wise convolutional layer, and a convolutional layer. As shown in Fig. 4, the input image is first processed by the feature extraction network to extract features, which are then divided into patches. These patches are then fed into the KAN block, which generates the predicted speckle noise. And the discriminator in SNA-SKAN is chosen as PatchGAN (Zhu et al., 2017).

3.6. Model training

The GAN loss function is defined as formula (3),

$$\text{LOSS}_{\text{GAN}} = \mathbb{E}[\log D(\text{image}_{\text{real}})] + \mathbb{E}[\log(1 - D(\text{image}_{\text{fake}}))] \quad (3)$$

In addition to the typical GAN loss, the generator loss also includes the supervised learning loss of three pairs of images, CI and CI_{BN_C} , CI and CI_{NG_C} , NI_C and $NI_C_{BN_C}$. These are computed using the L_1 loss function and are represented as (4), (5), (6).

$$\text{LOSS}_{G_1} = |CI - CI_{BN_C}| \quad (4)$$

$$\text{LOSS}_{G_2} = |CI - CI_{NG_C}| \quad (5)$$

$$\text{LOSS}_{G_3} = |NI_C - NI_C_{BN_C}| \quad (6)$$

Since LOSS_{G_1} and LOSS_{G_3} are calculated based on the base noise, it is necessary to adjust the weights to reduce the impact of the base noise on the overall loss. The total loss for this part is represented as (7).

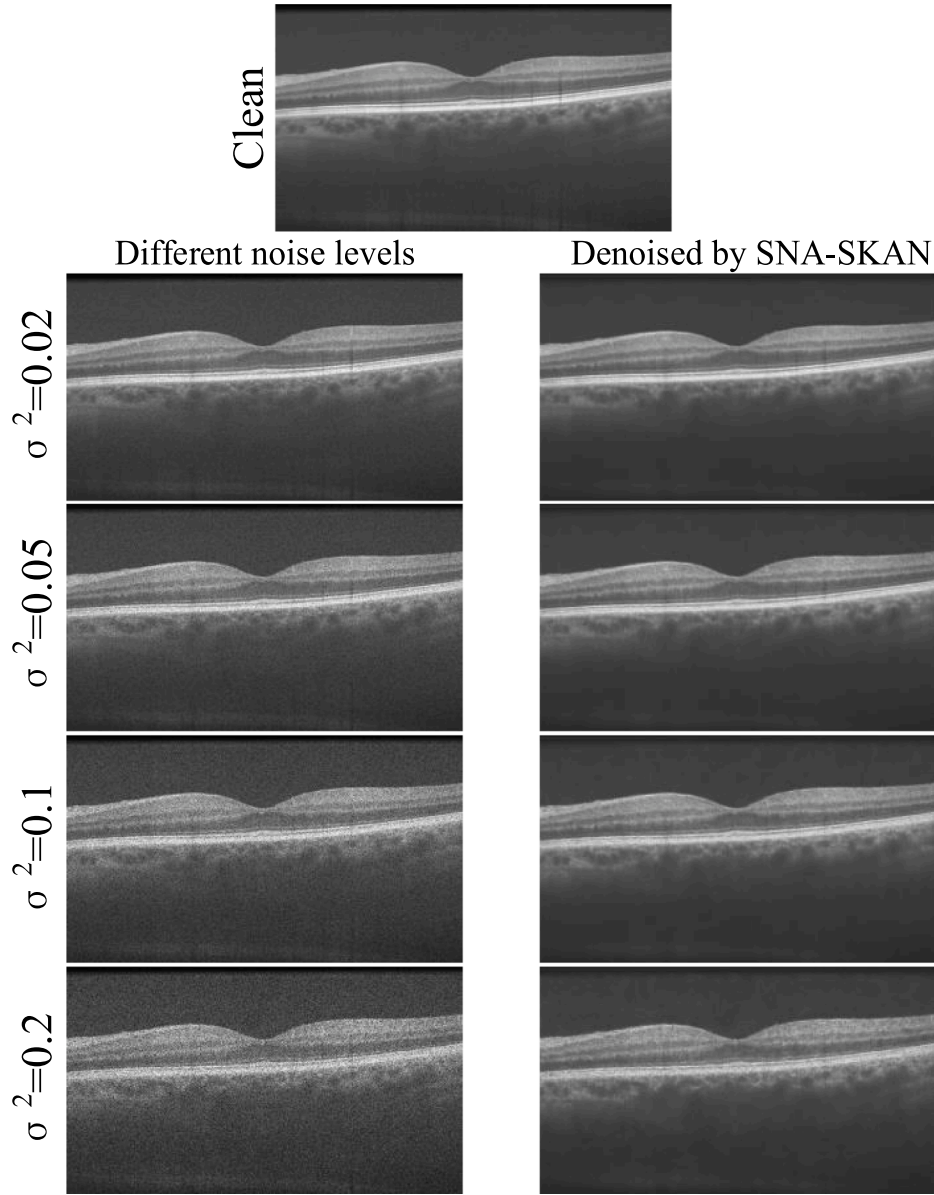


Fig. 7. The denoising results on simulated noisy OCT images. The left column displays simulated noisy OCT images at varying noise levels, while the right column displays the corresponding OCT images denoised by SNA-SKAN.

$$\text{LOSS}_G = \text{LOSS}_G_2 + \lambda(\text{LOSS}_G_1 + \text{LOSS}_G_3) \quad (7)$$

The training objective is that the generator can extract speckle noise from noisy OCT images. The entire training process of the method is outlined in [Algorithm 1](#).

Here, noise W denotes the noise extracted by wavelet transform, noise S denotes the noise extracted by SVD, base noise denotes the noise obtained by SNA. CI denotes the clean OCT image and NI represents the noisy OCT image, CI_{BN} denotes the noisy image obtained by sum of base noise CI . CI_{BN_C} denotes the predicted clean image obtained by subtracting $G(CI_{BN})$ from CI_{BN} . Noise G denotes the predicted noise outputted by the generator when applied to NI . NI_C denotes the predicted clean image obtained by subtracting noise G from NI . CI_{NG}

Table 6
Ablation results of SNA-SKAN.

Model	PSNR↑	SSIM↑	RMSE↓
Without SNA	26.102	0.761	0.050
Without SKAN	25.981	0.770	0.051
SNA-SKAN	26.651	0.814	0.046

denotes the noisy image obtained by sum of noise G and CI . NI_{C_BN} denotes the noisy image obtained by sum of base noise and NI_C . CI_{NG_C} denotes the predicted clean image obtained by subtracting $G(CI_{NG})$ from CI_{NG} . $NI_{C_BN_C}$ denotes the predicted clean image obtained by subtracting $G(NI_{C_BN})$ from NI_{C_BN} .

Algorithm 1. Training process

Data: Noisy OCT image NI ; clean OCT image CI

Neural Network: Generator G ; Discriminator D

Modules: SNA

Input noisy image NI to SNA, SNA outputs noise W , noise S and base noise.

For epoch:

$CI + \text{base noise} = CI_BN$

$CI_BN - G(CI_BN) = CI_BN_C$

Generate noise G : $G(NI) = \text{noise } G$

$NI - \text{noise } G = NI_C$

$CI + \text{noise } G = CI_NG$

$NI_C + \text{base noise} = NI_C_BN$

$CI_NG - G(CI_NG) = CI_NG_C$

$NI_C_BN - G(NI_C_BN) = NI_C_BN_C$

Calculate L_l LOSS: CI and CI_BN_C ; CI and CI_NG_C ; NI_C and $NI_C_BN_C$

Calculate G LOSS

Calculate D LOSS

End

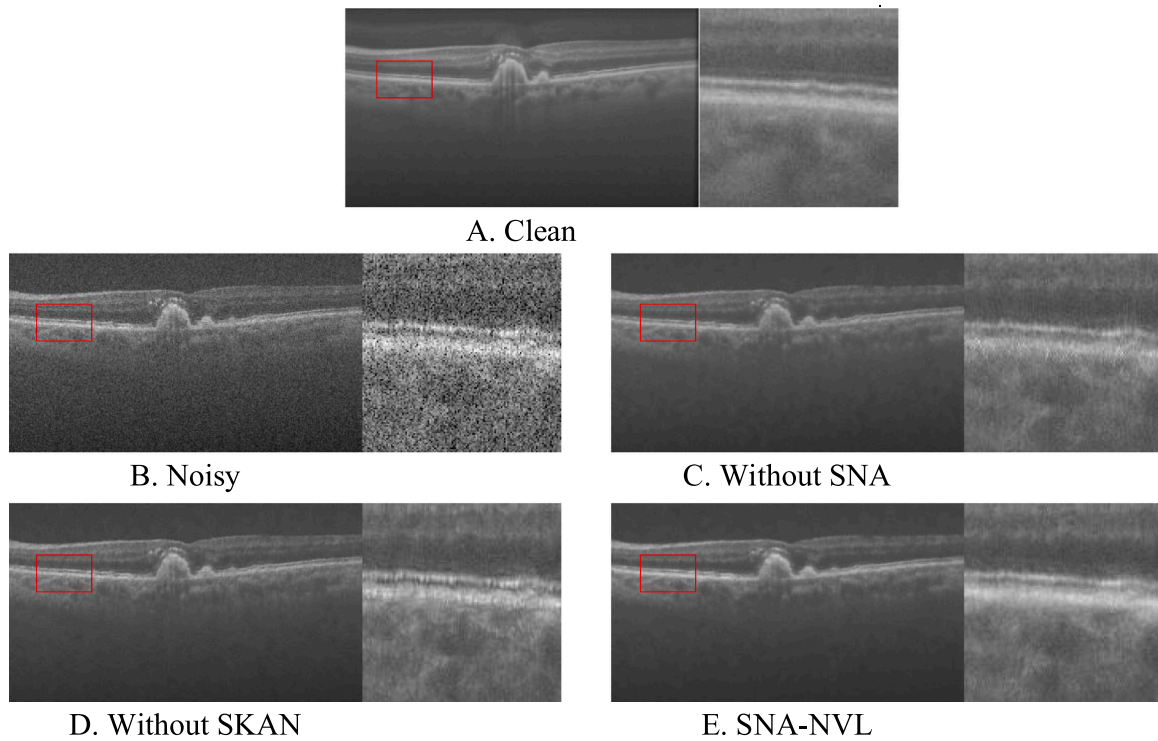


Fig. 8. The ablation results. (A) Clean OCT image; (B) Noisy OCT image; (C) Denoised result without the SNA module; (D) Denoised result without the SKAN module; (E) Denoised result using the full SNA-SKAN.

Table 7

Results of different λ .

The value of λ	PSNR $\uparrow$	SSIM $\uparrow$	RMSE $\downarrow$
1	26.435	0.810	0.048
0.5	26.651	0.814	0.046

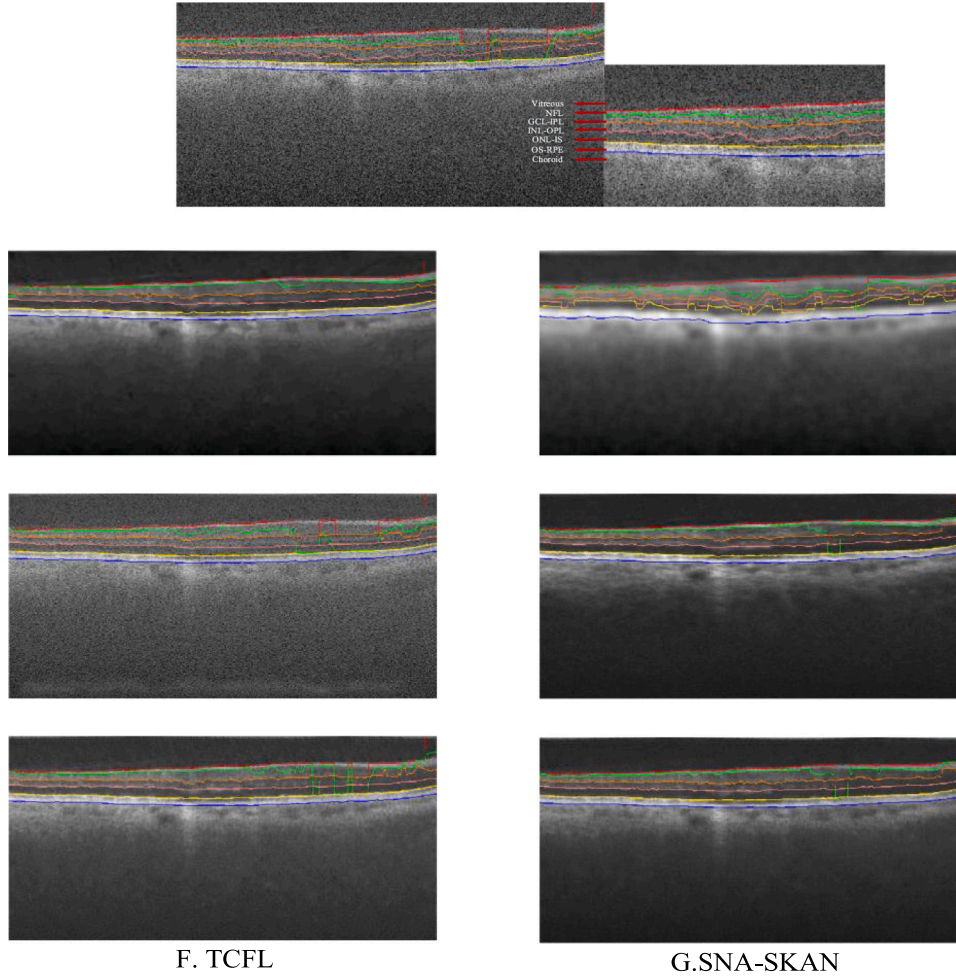


Fig. 9. Results on layer segmentation. (A) Noisy OCT image; (B) BM3D; (C) NLM; (D) DDPM; (E) CycleGAN; (F) TCFL; (G) SNA-SKAN.

4. Experimental results and analysis

4.1. Datasets and implementation details

SNA-SKAN is a deep learning-based unpaired denoising method that requires a relatively large amount of training data. PKU37 (Geng et al., 2022) is used to train the SNA-SKAN, consists of 1734 noisy OCT images and 37 clean OCT images. DUKE17 (Fang et al., 2012) and DUKE28 (Fang et al., 2013) are used as test datasets, containing 17 and 28 pairs of clean-noisy paired OCT images, respectively. The details of these three OCT datasets are shown in Table 1.

SNA-SKAN is implemented using the PyTorch framework, and all training is conducted on a single NVIDIA RTX 4090 GPU. Each noisy OCT image in PKU37 is paired with a clean OCT image to form an image pair used for training. The batch size is 2, Adam is adopted as the optimizer, the λ is 0.5, and the training epoch number is 100. The test datasets DUKE17 and DUKE28 are paired datasets, the selected evaluation metrics include peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), and root mean square error (RMSE). Higher PSNR and SSIM values, along with lower RMSE, indicate better image quality.

4.2. SNA-SKAN performance test

To evaluate the performance of SNA-SKAN, several representative models from different categories of denoising methods are selected for comparison. The selected models include BM3D (Danielyan et al., 2011), NLM (Buades et al., 2011), DDPM (Hu et al., 2022), CycleGAN (Gu et al.,

2021), and TCFL (Geng et al., 2022). BM3D and NLM are traditional denoising methods, DDPM is an unsupervised denoising method, CycleGAN is a classical unpaired denoising model, and TCFL is the state-of-the-art unpaired OCT denoising model.

BM3D and NLM are applied to the DUKE17 and DUKE28, with their parameters optimized through multiple empirical trials. DDPM is trained on all PKU37 noisy OCT images. CycleGAN is trained on the same training data as SNA-SKAN. For TCFL, the 1734 noisy OCT images are divided into 867 noisy image pairs, and one clean OCT image is appended to each pair to form a triplet. The experimental parameters for these three deep learning methods are consistent with their publicly available code. All models are evaluated on the DUKE17 and DUKE28 datasets after training. The test results are shown in Table 2 and 3, Fig. 5 and 6.

As shown in Table 2 and 3, SNA-SKAN significantly outperforms BM3D, NLM, DDPM, and CycleGAN across all evaluation metrics. Compared to TCFL, SNA-SKAN also demonstrates slightly superior performance. From Fig. 5 and 6, it can be seen that, the outer retinal layers of the clean OCT images exhibit clear delineation with clean background, and the inner retinal structures are well-defined. In contrast, the noisy OCT images show blurred outer retinal layers with significant background noise and less distinct inner retinal structures. BM3D and NLM reduce background noise to some extent, but result in substantial loss of image details. The outer retinal layers lack clarity, and the inner structures appear blurred. DDPM is an unsupervised method, which lacks a reference clean OCT during training. This leads to the introduction of additional noise with an unclear distribution, affecting the clarity of both inner and outer retinal layers. CycleGAN removes a large

portion of speckle noise but fails to preserve structural details in both inner and outer retinal layers. TCFL effectively suppresses speckle noise while maintaining structural integrity in both inner and outer retinal layers. Compared to the above methods, SNA-SKAN achieves more thorough speckle noise removal in both background and structural regions. Additionally, the detailed visualizations demonstrate that SNA-SKAN provides clearer delineation of the outer retinal layers. In summary, SNA-SKAN outperforms all compared models across all evaluation metrics and demonstrates superior capability in speckle noise suppression.

To validate the effectiveness of SNA-SKAN, a statistical significance analysis was conducted. Following the methodology in Feng et al. (2022), a paired Student's *t*-test was performed to evaluate the significance of the differences between TCFL and SNA-SKAN. Compared to TCFL, SNA-SKAN demonstrated statistically significant improvements, with *p*-values less than 0.001.

In addition, this paper analyzed model complexity by comparing the parameters and inference time of different methods. As shown in Table 4, DDPM requires the longest inference time among all methods due to its iterative denoising process, while the two traditional methods also exhibit relatively long inference times. In contrast, SNA-SKAN introduces only a slight increase in parameter count and inference time compared to TCFL, remaining well within an acceptable range for clinical practice.

4.3. Robustness analysis

To further evaluate the denoising capability of SNA-SKAN under varying speckle noise intensities, synthetic speckle noise of different levels is added to clean OCT images from DUKE17. The simulated noise is characterized by its variance σ^2 , four variance levels ($\sigma^2 = 0.02, 0.05, 0.1, 0.2$) are selected, each level corresponds to one specific intensity of speckle noise, larger σ^2 indicates stronger noise intensity. These simulated noisy OCT images are denoised using SNA-SKAN trained on the PKU37. The denoising results under different noise intensities are shown in Table 5 and Fig. 7. From Fig. 7, it can be observed that the simulated noisy OCT closely resembles the clean OCT when the intensity is 0.02, and the denoised OCT is also very close to the clean OCT. As the noise intensity increases, the quality of the simulated noisy OCT images progressively degrades. At the highest noise level ($\sigma^2 = 0.2$), the simulated noise severely obscures structural details in the OCT images. Nevertheless, the denoised OCT images not only remove a substantial amount of noise but also retain most structural details in both inner and outer retinal layers. These results demonstrate that SNA-SKAN is robust to varying levels of speckle noise.

4.4. Ablation study

The purpose of the ablation study is to evaluate the individual contributions of the SNA and SKAN modules to the overall performance of the proposed denoising model. The ablation experiments are conducted using PKU37 for training and DUKE17 for testing. For SNA, because it requires two types of noise as input, two different intensities (σ^2 is 0.05 and 0.15) of simulated speckle noise are selected as input. For SKAN, DNCNN is chosen as the noise extraction network. The ablation results are shown in Table 6 and Fig. 8. As shown in Table 6, removing either the SNA or SKAN module results in a consistent decline across all evaluation metrics. Fig. 8 further illustrates that the absence of SNA leads to reduced structural clarity in the denoised OCT images. This is because the speckle noise extracted by SNA from real noisy OCT images is more reliable than simulated speckle noise. And the absence of SKAN results in noticeable background noise, which can be attributed to the superior ability of the KAN-based model to capture complex noise distributions compared to general noise generation models. These results demonstrate that both the SNA and SKAN modules play essential roles in enhancing the denoising performance of SNA-SKAN.

Additionally, this paper explores the effect of different λ in formula (7). Table 7 shows the results of different λ . It can be observed that λ sets to 0.5 performs better than λ sets to 1. Considering that two part losses are calculated based on base noise, the λ sets to 0.5 can balance the effects of base noise and noise G .

4.5. Application in layer segmentation

OCT denoising can provide clinicians with a better visual experience and can also be used as image preprocessing for high-level vision tasks. Layer segmentation of OCT images is very important for the diagnosis and research of eye diseases (He et al., 2021). Therefore, this paper selects the 2011 IOVS Chiu dataset (Chiu et al., 2012) to verify the effectiveness of OCT denoising through its application to layer segmentation. This dataset was proposed by Duke University and obtained using an SDOCT system. The image resolution is 512×1000 pixels. The images are processed using different denoising algorithms and then fed into OCTSEG, a public segmentation tool. OCTSEG divides each image into seven layers, from top to bottom: Vitreous, NFL, GCL+IPL, INL+OPL, ONL+IS, OS+RPE, and Choroid. As shown in Fig. 9, the segmentation results improve significantly compared to those obtained from the original noisy OCT images. SNA-SKAN achieves the most significant improvement. Although some segmentation errors remain, the overall enhancement greatly reduces the need for manual correction.

4.6. Discussion

For denoising methods, preserving image details and effectively removing noise are equally important. Traditional denoising methods often struggle to achieve both simultaneously and typically require iterative parameter tuning to reach optimal performance. As a result, they are seldom adopted for solving complex noise problems in modern imaging tasks. Instead, deep learning-based approaches have become the preferred choice due to their superior performance and automation. However, traditional methods can still yield unexpected results when used as assisting methods to enhance the performance of neural network. The key focus of deep learning-based methods is to train the most effective neural network. Designing training algorithms or altering network architectures to improve denoising performance, such as CycleGAN, DDPM-based methods, TCFL, and the SNA-SKAN proposed in this paper, all originate from method aimed at training neural networks. For such methods, the focus is on reducing the demand for training data, simplifying the model framework, and ensuring the stability of the training process. Compared to CycleGAN-based methods that utilize two generators and two discriminators, SNA-SKAN adopts a single generator-discriminator architecture, thereby simplifying the training process. And SNA-SKAN does not require the long inference time typically associated with DDPM-based methods. Moreover, unlike TCFL, which requires triplet datasets comprising two noisy images and one clean image, SNA-SKAN operates on unpaired clean and noisy images, significantly lowering the dataset requirements. Nevertheless, SNA-SKAN has its limitations. This method is based on the premise that speckle noise in OCT in the logarithmic domain is additive noise. It can be observed that in the noisy OCT images, the noise appears relatively uniform. However, in modalities like low-dose CT, the noise is often a complex mixture of Gaussian and Poisson distributions with spatially varying intensity. In such cases, the current method may not be directly applicable. For denoising tasks involving additive speckle noise, such as in ultrasound imaging, where noise arises from tissue scattering, SNA-SKAN demonstrates strong potential for clinical application, particularly in scenarios lacking paired clean-noisy datasets.

5. Conclusion

The proposed SNA-SKAN addresses the problem of OCT denoising in scenarios where clean and noisy images are unpaired. SNA-SKAN is

based on a GAN framework consisting of a single generator and a single discriminator. The SNA module leverages intrinsic information from noisy OCT images to extract high-confidence speckle noise. To prevent the extracted noise from being fixed, the structural information it contains is disrupted using patch shuffle, which in turn guides the training of the neural network. SKAN leverages KAN's capacity to fit complex distributions, thereby enhancing its ability to generate realistic speckle noise. Extensive experiments are conducted to evaluate the denoising performance of SNA-SKAN. The results demonstrate that SNA-SKAN outperforms existing unsupervised and state-of-the-art unpaired methods in terms of both denoising effectiveness and structural preservation.

CRediT authorship contribution statement

jiang zhencun: Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **ren kangrui:** Visualization, Validation. **hao zixiong:** Visualization, Formal analysis, Data curation. **wang zhongjie:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Zhongjie wang reports administrative support, equipment, drugs, or supplies, and statistical analysis were provided by Tongji University. Zhongjie wang reports a relationship with Tongji University that includes: employment. There are no conflicts of interest. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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